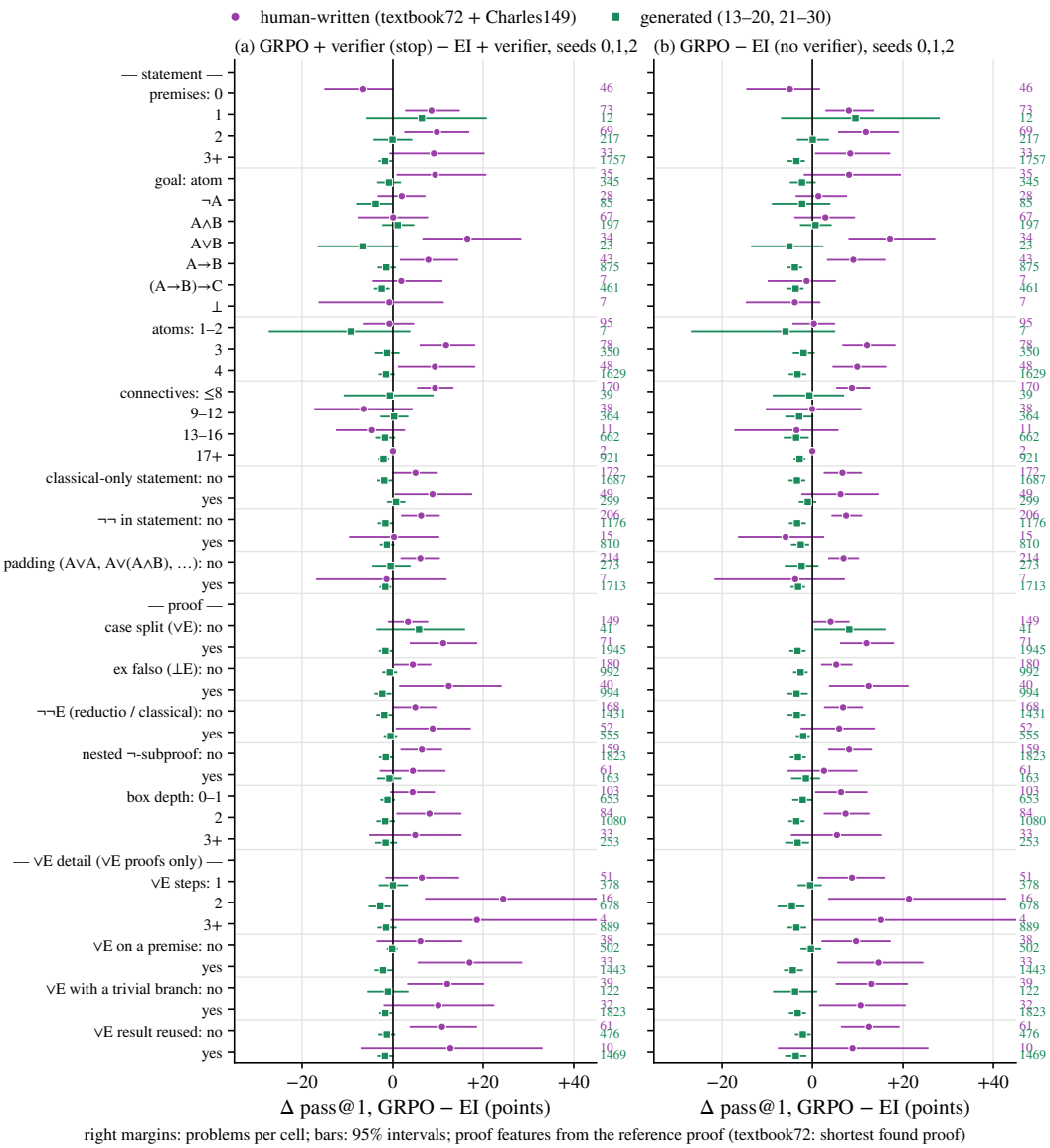
human-written (textbook72 + Charles149)
generated (13–20, 21–30)
(a) GRPO + verifier (stop) − EI + verifier, seeds 0,1,2
(b) GRPO − EI (no verifier), seeds 0,1,2
— statement —
premises: 0
1
2
3+
goal: atom
¬A
A∧B
A∨B
A→B
(A→B)→C
⊥
atoms: 1–2
3
4
connectives: ≤8
9–12
13–16
17+
classical-only statement: no
yes
¬¬ in statement: no
yes
padding (A∨A, A∨(A∧B), …): no
yes
— proof —
case split (∨E): no
yes
ex falso (⊥E): no
yes
¬¬E (reductio / classical): no
yes
nested ¬-subproof: no
yes
box depth: 0–1
2
3+
— ∨E detail (∨E proofs only) —
∨E steps: 1
2
3+
∨E on a premise: no
yes
∨E with a trivial branch: no
yes
∨E result reused: no
yes
−20
0
+20
+40
Δ pass@1, GRPO − EI (points)
right margins: problems per cell; bars: 95% intervals; proof features from the reference proof (textbook72: shortest found proof)